

Instrumental Variables

Chris Conlon

Tuesday 21st October, 2025

Applied Econometrics

► Basic Idea of Instrumental Variable (IV):

- What if we have a variable that is correlated with X *but not with* Y
- Then any changes in Y caused by that variable will reflect causal changes by X

► Equation of interest:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

► IVs break X into two pieces that are themselves uncorrelated:

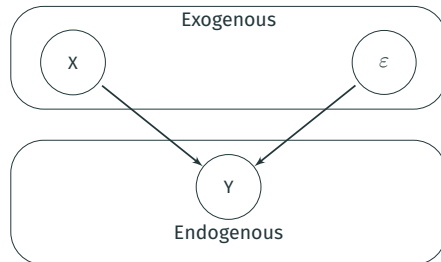
$$X_i = \gamma_0 + \gamma_1 Z_i + \eta_i$$

- A piece that is not correlated with ε ($Cov(Z, \varepsilon) = 0$)
- A piece that is correlated with ε ($Cov(\eta, \varepsilon) \neq 0$) – source of the endogeneity problem
- Finally, $Cov(Z, \eta) = 0$

► Z_i is an **instrumental variable**

Terminology Review

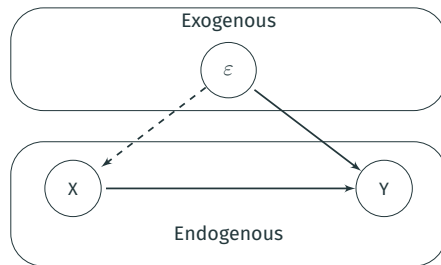
A “Good” Regression:



- ▶ **Exogenous Variables:** Variables in the data that do not cause each other
 - ε is always exogenous, so exogenous also just means variables not correlated with ε
- ▶ **Endogenous Variables:** Variables that are determined by exogenous variables in the model
 - ε is always in Y so Y is always endogenous

Omitted Variable Bias with Pictures

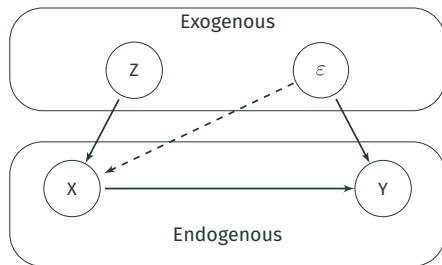
Selection/OVB:



- ▶ In this picture X is **endogenous** because ϵ now causes X as well
- ▶ What if there is another exogenous variable that *does not* directly cause Y ?

Omitted Variable Bias with Pictures

Instrumental Variable:



- ▶ Z causes Y only indirectly through X
- ▶ We can estimate "causal effect" of Z on Y and this MUST be the causal effect of Z on X and the causal effect of X on Y
 - Mathematically we need to split effect of Z on Y into effect of Z on X and X on Y
 - Related concept in statistics: Path Analysis

Formal Definition of an Instrumental Variable

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

► We call a variable Z a valid **instrumental variable** if the following two conditions hold:

1. **Relevance:** $Cov(X, Z) \neq 0$

- An arrow from Z to X in the pictures

2. **Exogeneity:** $Cov(\varepsilon, Z) = 0$

- No arrow from Z to Y or Z to ε in the picture

Key Assumption #1 of Instrumental Variables

- ▶ **Relevance:** $Cov(X, Z) \neq 0$
- ▶ This assumption just means that X and Z are correlated
- ▶ We observe both X and Z , so can easily test this assumption by regressing:

$$X_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$$

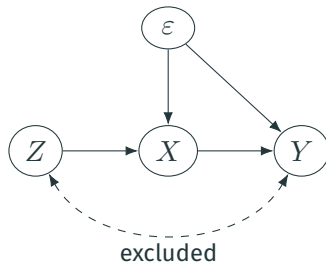
- ▶ If $\beta_1 \neq 0$ in regression, we say instrument is relevant

Key Assumptions #2 of Instrumental Variables

- ▶ **Exogeneity (or validity):** $Cov(\varepsilon, Z) = 0$
- ▶ This assumption means that Z and ε cannot be correlated
- ▶ We do not observe ε , so we **cannot** test this assumption
- ▶ In general, we need to “defend” this assumption by telling a story about why Z and ε are unlikely to be correlated
- ▶ Much like parallel trends or strict exogeneity, these crucial identifying assumptions cannot be tested on their own. However, we can test them against each other. That is, the best we can do is ask the question “given this exogeneity assumption, does this other exogeneity assumption seem true?” But we can never test an exogeneity assumption in a self-contained way.

IV: Causal Diagram

- ▶ Recall the absence of direct path between $Z \rightarrow Y$. This is the **exclusion restriction**.
- ▶ $Z \rightarrow X$ is the **relevance** condition.



Idea: There is a path from $Z \rightarrow Y$ but only **through** X (this is the key).

Best Defense of Exogeneity IV Assumption: Randomized Experiment

- ▶ Back to the class size example:

$$score_i = \beta_0 + \beta_1 CS_i + \varepsilon_i$$

- ▶ where:

- $score_i$: Test score of student i
- CS_i : Class size of student i

- ▶ Suppose that we use a coin flip that sends kids that get a “head” to a small class and kids getting a “tails” to big class
 - This is our randomized experiment!
- ▶ Let’s call the coin flip our instrument Z (where $Z_i = 1$ if heads, $Z_i = 0$ if tails)

Best Defense of Exogeneity IV Assumption: Randomized Experiment

$$score_i = \beta_0 + \beta_1 CS_i + \varepsilon_i$$

- ▶ Is Z (our coin flip) a good instrument?
- ▶ **Relevance:** $Cov(CS, Z) \neq 0$? Yes, if $Z_i = 1$ kid gets small class, if $Z_i = 0$ kid gets big class
 - So the regression $CS_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$ will estimate that $\beta_1 < 0$
- ▶ **Exogeneity:** $Cov(\varepsilon, Z) \neq 0$? Untestable – so need to tell a story
 - Story: Exogeneity holds because coin flip is random and does not depend on any student or parent characteristics that would affect test scores. Therefore, there is nothing related to Z (besides X) that is also related to test scores, so ε and Z must be uncorrelated.
This is a good story! It's the “magic” of randomization.

Randomized Experiment as an IV

- ▶ So a randomized experiment can be treated as an IV, but there is a big difference in how we evaluate the assumption for an experimental or non-experimental setting
- ▶ When running an experiment, the exogeneity assumption is justified if the randomization was implemented in way that was not correlated with anything else that could influence outcomes. Attrition and manipulation can undermine exogeneity in an experimental context if they create a correlation between treatment status and other factors that might influence outcomes. Thus, exogeneity amounts to an assumption about how the randomization was implemented.
- ▶ In contrast, the IV exogeneity assumption in a non-experimental context is a broad and vague statement about how the world works; it's much more difficult to interpret and evaluate. The same is true of strict exogeneity or parallel trends.

IV Estimation: Example with a binary $Z \in \{0, 1\}$

Let's begin with the simplest 0/1 model (this time for Z):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ X is endogenous but we have a 0/1 instrument, Z
 - Because Z is 0/1 we can focus on group means.
- ▶ What assumptions do we need to make?
 - $Cov(X, Z) \neq 0$
 - $Cov(Z, \varepsilon) = 0$
- ▶ The intuition from above: solve for the “indirect” effect of Z on Y

The Relationship Between Y on Z

What is the relationship between Y and Z , taking expectations?

$$\begin{aligned}\mathbb{E}(Y|Z) &= \mathbb{E}(\beta_0 + \beta_1 X + \varepsilon|Z) \\ &= \beta_0 + \beta_1 \mathbb{E}(X|Z) + \underbrace{\mathbb{E}(\varepsilon|Z)}_{=0} \\ &= \beta_0 + \beta_1 \mathbb{E}(X|Z)\end{aligned}$$

Since Z is 0/1...

$$\begin{aligned}\mathbb{E}(Y|Z = 1) &= \beta_0 + \beta_1 \mathbb{E}(X|Z = 1) \\ \mathbb{E}(Y|Z = 0) &= \beta_0 + \beta_1 \mathbb{E}(X|Z = 0)\end{aligned}$$

Rearranging...

$$\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0) = \beta_1 \times (\mathbb{E}(X|Z = 1) - \mathbb{E}(X|Z = 0))$$

The IV Regression for a binary $Z \in \{0, 1\}$

From the previous slide we have:

$$\beta_1 = \frac{\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0)}{\mathbb{E}(X|Z = 1) - \mathbb{E}(X|Z = 0)}$$

From the LLN we can construct the following estimator:

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- ▶ The subscript refers to $Z = 1$ or $Z = 0$
- ▶ We call this the IV estimator

Interpreting the Estimator for binary $Z \in \{0, 1\}$

Estimator:

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- For a binary X we look at the change in Y over the change in X :

$$\hat{\beta}_1^{OLS} = \frac{\bar{Y}_{X=1} - \bar{Y}_{X=0}}{\bar{X}_{X=1} - \bar{X}_{X=0}} = \Delta \bar{Y}$$

- For a 0/1 X , $\Delta \bar{X} = 1$
- Basically looking at how much Y changes for a change in X

- Intuition for a 0/1 Z

- The effect of X on Y is still the change in Y over a change in X
- Use the change in Y *given* Z to shut down the effects of ε
- Use the change in X *given* Z to get the right scaling

The IV Regression with a Continuous Z

Same model as before:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ Now assume that Z is continuous
- ▶ How to capture the indirect effect of Z ?
 - Intuition from OLS: use the covariance between Z and Y as a measure of the relationship:

$$\begin{aligned} \text{Cov}(Y, Z) &= \text{Cov}(\beta_0 + \beta_1 X + \varepsilon, Z) \\ &= \beta_1 \text{Cov}(X, Z) \end{aligned}$$

- Rearranging:

$$\beta_1 = \frac{\text{Cov}(Y, Z)}{\text{Cov}(X, Z)}$$

- ▶ **Instrumental Variables Estimator:** $\hat{\beta}_1^{IV} = s_{YZ}/s_{XZ}$

Heterogeneous treatment effects I

- ▶ Let's consider a simple case with binary treatment $X_i \in \{0, 1\}$ where the treatment effect $\beta_i = \mathbb{E}(Y_i(1) - Y_i(0))$ is heterogeneous.
 - $Y_i(x)$ denotes the potential outcome for i given treatment status $X_i = x$.
- ▶ We have a binary instrument $Z_i \in \{0, 1\}$
- ▶ Let's also have potential outcomes for treatment status: $X_i(z)$ is the potential treatment status for i when $Z_i = z$.
- ▶ Model:

$$Y_i = \beta_0 + \beta_i X_i + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_i X_i + \varepsilon_i$$

► Assuming that $\mathbb{E}[\varepsilon_i | Z_i = 1] - \mathbb{E}[\varepsilon_i | Z_i = 0] = 0$,

$$\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0] = \mathbb{E}[\beta_i X_i | Z_i = 1] - \mathbb{E}[\beta_i X_i | Z_i = 0]$$

Plugging in our potential outcomes for X ,

$$\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0] = \mathbb{E}[\beta_i (X_{i1} - X_{i0})]$$

$$\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0] = \mathbb{E}[\beta_i(X_{i1} - X_{i0})]$$

- Given the binary treatment, there are three possible values for $X_{i1} - X_{i0}$:

$$\begin{aligned}\mathbb{E}[\beta_i(X_{i1} - X_{i0})] &= \Pr(X_{i1} - X_{i0} = 1) \mathbb{E}[\beta_i(X_{i1} - X_{i0}) | X_{i1} - X_{i0} = 1] && \text{(compliers)} \\ &+ \Pr(X_{i1} - X_{i0} = 0) \mathbb{E}[\beta_i(X_{i1} - X_{i0}) | X_{i1} - X_{i0} = 0] && \text{(always/never takers)} \\ &+ \Pr(X_{i1} - X_{i0} = -1) \mathbb{E}[\beta_i(X_{i1} - X_{i0}) | X_{i1} - X_{i0} = -1] && \text{(defiers)}\end{aligned}$$

- The second term (always/never takers) is necessarily equal to zero. We might also make a **monotonicity assumption** that $\Pr(X_{i1} - X_{i0} = -1) = 0$.
- Ruling out defiers, this could be interpreted to mean that being assigned to treatment can only make you more likely to receive treatment.

Heterogeneous treatment effects IV

- Start with

$$\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0] = \mathbb{E}[\beta_i (X_{i1} - X_{i0})]$$

- **Monotonicity** means:

$$\mathbb{E}[\beta_i (X_{i1} - X_{i0})] = \Pr(X_{i1} - X_{i0} = 1) \mathbb{E}[\beta_i (X_{i1} - X_{i0}) | X_{i1} - X_{i0} = 1]$$

- Also note that $\Pr(X_{i1} - X_{i0} = 1) = E[X_i | Z_i = 1] - E[X_i | Z_i = 0]$

- Combining the above,

$$\begin{aligned} \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[X_i | Z_i = 1] - \mathbb{E}[X_i | Z_i = 0]} &= \mathbb{E}[\beta_i (X_{i1} - X_{i0}) | X_{i1} - X_{i0} = 1] \\ &= \mathbb{E}[\beta_i | X_{i1} - X_{i0} = 1] \end{aligned}$$

- We now have that our IV estimator equals the **local average treatment effect**, which means the average effect on the individuals whose treatment status is changed by the instrument.

IV Example

- Suppose that we are interested in investigating the effect of studying on grades:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- where

- Y_i : is GPA
- X_i : is study time (hours per day)

- We expect that our OLS estimator $\hat{\beta}_1$ will be severely biased here (why?)

- ▶ To get rid of OVB, we shall use an IV
- ▶ One possible IV: whether your roommate has a N64 (or playstation/whatever video game console kids use today)
- ▶ Important feature: Roommates are randomly assigned in college
 - At least at Berea College (Kentucky) where this example comes from (see Stinebrickner and Stinebrickner (2008))

- ▶ First thing for any instrument is to think of validity. Two conditions:
 - Instrument is $N64$, which is indicator variable that your roommate has N64
- 1. Relevance: $Cov(Z, X) \neq 0$; here $Cov(N64, study) \neq 0$
 - Seems likely to hold as everyone prefers playing Mario Kart to studying
 - Testable
- 2. Exogeneity: $Cov(Z, \varepsilon) = 0$; here $Cov(N64, \varepsilon) = 0$
 - Untestable
 - Is it plausible?

Thinking About Exogeneity

- ▶ Exogeneity: $Cov(Z, \varepsilon) = 0$
- ▶ “Storytime” should talk about how two things hold
 1. People select into Z in some manner that is likely to be correlated with Y
 - Concern: People that pick roommates that are “fun” and have N64s are likely people who do not care too much about grades
 2. Z only affects Y through X
 - Concern: Having a roommate with a video game affects your GPA through other means than affecting your study hours
- ▶ Either story “invalidates” the instrument (but in different ways)

1. Selection story: there is an omitted variable related to both Z and Y
 - Example: omitted variable is effort because students that really put in a lot of effort make sure they do not get a roommate with N64
 2. Other channel story:
 - Possible Story 1: Mario Kart helps me grasp physics, so I ace my physics exam
 - so Z directly affects Y *independent of X*
 - Possible Story 2: Other people hang out in our room due to our N64, making it really loud and so I cannot study effectively
 - so Z affects Y through *another X*
- Conceptually, there are two aspects to the exogeneity assumption: **randomization** and **exclusion**. Be careful: even when an IV is clearly random, it might not be excludable.

IV Math

IV Directly into Model (aka “the reduced form”)

- ▶ Suppose that our IV assumptions hold
- ▶ Then we can just directly replace X with our IV in our model:

$$Y_i = \pi_0 + \pi_1 Z_i + \varepsilon_i$$

- ▶ where
 - Y_i : is GPA
 - Z_i : is whether roommate has N64
- ▶ **Interpretation of $\hat{\beta}_1$** : Having a roommate with a N64 **causes** a $\hat{\beta}_1$ decrease in GPA
 - Causal effect because validity of instrument, $cov(Z, \varepsilon) = 0$, implies OLS is consistent for above equation. Of course we could question the instrument's validity.
- ▶ But we want the effect of *studying* on GPA!

Two Stage Least Squares

A popular IV estimator is Two Stage Least Squares (2SLS)

- ▶ This effectively runs regression $Y_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$, but scales β_1 by how much Z affects X
 - If Z affects X at one-to-one rate, β_1 in above regression will capture effect of X
 - If Z affects X only a little, we need to scale β_1 up a lot to say what effect of X on Y is
- ▶ We therefore estimate in two steps:
 1. Regress X on Z to capture effect of Z on X
 2. Regress fitted values of step 1 to capture effect of X on Y
- ▶ Under IV assumptions, this gives us **causal** effect of X on Y

- ▶ Step 1: regress

$$X_i = \pi_0 + \pi_1 Z_i + \eta_i$$

- ▶ Step 2: take your predicted values from Step 1, $\hat{X}_i$, and regress

$$Y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$$

- ▶ Under IV assumptions, step 1 finds the “good part” of X (that is not related to ε), and then step 2 takes that “good part” of X as a regression to find the causal relationship between X and Y

► Why is $\hat{\beta}_1^{2SLS} = \frac{Cov(Y, Z)}{Cov(X, Z)}$?

► True model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

► Take covariance of both sides with respect to Z :

$$Cov(Y_i, Z_i) = Cov(\beta_0 + \beta_1 X_i + \varepsilon_i, Z_i)$$

$$Cov(Y_i, Z_i) = Cov(\beta_0, Z_i) + Cov(\beta_1 X_i, Z_i) + Cov(\varepsilon_i, Z_i)$$

$$Cov(Y_i, Z_i) = 0 + \beta_1 Cov(X_i, Z_i) + 0 \text{ since } Cov(\varepsilon_i, Z_i) \text{ by assumption}$$

$$\implies \beta_1 = \frac{Cov(Y, Z)}{Cov(X, Z)}$$

► Data:

- Y_i : is GPA
- X_i : is study time (hours per day)
- Z_i : is indicator for roommate having N64

► Running 2SLS by doing both steps manually:

► Step 1: Regress $X_i = \pi_0 + \pi_1 Z_i + v_i$ and get predicted study time, $\hat{X}_i$

- `Data$xHat = predict(feols(X~Z, data=Data))`

► Step 2: Regress $y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$

- `m3 = feols(Y~xHat, data=Data)`
- `summary(m3, vcov= "HC1")`

■ Note: Standard errors will be wrong!

- `predict` gives the fitted values of X from the first stage
- Then `feols` can be used (again) to get the second stage
- Problem: the standard errors (even robust) will be wrong

R can run 2SLS all at once:

- ▶ `m4 = feols(Y ~ W | X ~ Z, data=Data)`
- ▶ First formula: Regress Y on **exogenous controls** W (and endogenous X)
- ▶ Second formula: Write **first-stage** regression for **endogenous** X (included in second stage automatically)
- ▶ `feols` will also adjust standard errors

This is basically why we use `feols` whenever we can.

IV Setup with Matrix Notation

- ▶ Consider a multiple regression equation

$$y = \beta X + \varepsilon$$

The IV assumptions are now that

- $Cov(z_i, \varepsilon_i) = 0$
 - $Cov(z_i, x_i)$ has *full rank*, i.e., rank K , where K is the number of regressors in $\mathbf{x}$
- ▶ We can have multiple regressors as well as instruments. Some regressors can also be instruments; if $z_i = x_i$ then we can just use OLS.
 - ▶ Regressors that are instruments are **exogenous regressors**. Regressors that are not instruments are **endogenous regressors**.
 - ▶ For the rank assumption, we need at least one instrument for each endogenous regressor.

- The 2SLS formula is now

$$b_{2SLS} = \left(\hat{X}' \hat{X} \right)^{-1} \hat{X}' y$$

with $\hat{X} = Z \left(Z' Z \right)^{-1} Z' X$

- With homoscedasticity, the asymptotic variance of the estimator is

$$Var(b_{2SLS}) = \frac{s^2}{n} \left[Z' X (X' X)^{-1} X' Z \right]^{-1}$$

- Note: this (asymptotically correct) estimate of the variance is not equal to the (incorrect) formula we would get from naïvely applying OLS formulas to the second-stage regression.

- ▶ Before the $|: Y \sim Controls$ (endogenous X is automatically included)
- ▶ After the $|: X \sim Z$ (excluded instruments only, the exogenous controls are included automatically)
- ▶ Syntax differs for different packages → Be careful!

```
library(fixest)
library(tidyverse)
library(ddml)
library(car)

babydata <- data.frame(AE98) %>%
  mutate(BoyBoy = 1* (boy1st ==1 & boy2nd==1),
         GirlGirl = 1* (boy1st ==0 & boy2nd==0))

m5 = feols(workedm | morekids + boy1st + boy2nd + black + hispan +
  othrace + agem1 + agefstm | same-sex ~ boy1st + boy2nd,
  data=babyData)
```

Comparing 2SLS Variance to OLS Variance

For the single-variable case, we can show that

$$Var(\hat{\beta}_{2SLS}) = \frac{1}{N} \frac{\sigma_{\varepsilon}^2}{\sigma_X^2 R_{XZ}^2} \mathbf{x}' \mathbf{e}_{OLS} = \mathbf{0}$$

OLS variance under homoscedasticity: $Var(\hat{\beta}^{OLS}) = \frac{1}{N} \frac{\sigma_{\varepsilon}^2}{\sigma_X^2}$

- ▶ Since $R_{XZ}^2 \leq 1$, IV variance is larger
- ▶ The higher first stage R_{XZ}^2 , the lower the variance. If X and Z are collinear, variance is equivalent to OLS.
- ▶ If $R_{XZ}^2 \approx 0$ then variance blows up!

$$\begin{aligned}\mathbb{E}[\hat{\beta}_1^{2SLS}] &= \mathbb{E}\left[\frac{\sum_{i=1}^N (z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})(z_i - \bar{z})}\right] \\ &= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^N (z_i - \bar{z})\varepsilon_i}{\sum_{i=1}^N (x_i - \bar{x})(z_i - \bar{z})}\right] = \beta_1 + 0\end{aligned}$$

- ▶ The two major assumptions:
 1. Relevance: $Cov(X, Z) \neq 0$, or that $\mathbb{E} [\mathbf{z}_i \mathbf{x}_i']$ has full rank
 2. Exogeneity: All instruments must be uncorrelated with ε
- ▶ First, let's talk about the relevance assumption.
- ▶ It's easy to check that the sample analog to $Cov(X, Z) \neq 0$, or the analogous matrix condition.
- ▶ The more practically relevant concern is that the covariance will be *close* to zero. This is called **weak instruments**.
- ▶ How do we test and deal with weak instruments in practice?

- ▶ In finite samples there's a possibility that the sample covariance

$$\sum_{i=1}^N (x_i - \bar{x})(z_i - \bar{z})$$

is close to zero, especially if the true covariance is relatively small.

- ▶ Dividing by zero is *bad*.
- ▶ Note we don't have the same problem in OLS as long as we have variation in the regressors.

- ▶ There is **big** problem with IV estimation: 2SLS performs **terribly** in small samples
- ▶ Intuition: Instrumental variables relies on $Corr(X, Z) \neq 0$
 - If $Cov(X, Z) = 0$ then IV estimator is infinite!
- ▶ In practice, very rarely will $Cov(X, Z) = 0$ in data (just randomness)
 - Hence, IV will usually “work” with any choice of X and Z
 - Realistically, sometimes $Cov(X, Z)$ is small if Z only explains a small portion of X
- ▶ If $Cov(X, Z)$ is small we say that Z is a weak instrument

Weak Instruments and Small Sample Bias

Suppose that Z is a *bad instrument* in that $Cov(Z, \varepsilon)$ are correlated

- ▶ Similar to OVB

$$\hat{\beta}_1^{IV} \rightarrow \beta_1 + \frac{Cov(Z, \varepsilon)}{Cov(Z, X)} \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- ▶ Compare to OLS case (biased because of OVB):

$$\hat{\beta}_1^{OLS} \rightarrow \beta_1 + Cov(X, \varepsilon) \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- ▶ Which estimator is better here?

- IV's advantage is $Corr(Z, \varepsilon)$ should be smaller than $Corr(X, \varepsilon)$
- But if instrument is weak $Corr(Z, X) \approx 0$, then bias blows up even if $Corr(Z, \varepsilon)$ small

- ▶ Danger of weak instruments:

- Very small bias in Z can be greatly magnified if instruments are weak
- Amplified bias may have different sign than OLS

To do inference we rely on the CLT:

$$\hat{\beta}^{IV} \sim \mathcal{N}(\beta, \text{Var}(\hat{\beta}^{IV}))$$

- ▶ First issue: the Normal Approximation fails unless N is huge! (the CLT does not work well)
 - Intuition: $\hat{\beta}^{IV} = \text{Cov}(Y, Z) / \text{Cov}(X, Z)$.
 - If $\text{Cov}(X, Z) \approx 0$ then small changes in estimated covariance can have huge effects on estimated coefficients

- ▶ Second issue: even if the normal approximation works the variance explodes!
 - Intuition from homoscedastic case: $\text{Var}(\hat{\beta}^{IV}) = \sigma^2 / (N\sigma_X^2 R_{XZ}^2)$
 - If the R^2 of X on Z is very small, then variance is very large

How To Deal With Weak Instruments?

What are we to do if instruments are weak?

- ▶ Easiest: get more instruments or better instruments
- ▶ Harder: adjust how to do inference
 - Turns out that if instruments are weak, asymptotic distribution of $\hat{\beta}$ will be the *ratio* of two *correlated* normal random variables
 - We will not pursue this (uses LIML)
- ▶ The good news: We can test $Cov(X, Z)$ by regressing $x_i = \pi_0 + \pi_1 Z_i + \eta_i$
 - Since Z and X are data, we can just see if $\hat{\pi}_1$ is far from zero
- ▶ **Rule of Thumb:** Only use instrument if t-stat of null hypothesis that $\pi_1 = 0$ is ≥ 10 (much bigger than stat significance of 1.96!)

Some clarification:

- ▶ With only one endogenous variable, the $F > 10$ cutoff refers to the first-stage regression of that endogenous variable on all the exogenous variables. The null hypothesis for that F-statistic is that the coefficients on all the excluded instrumental variables are zero.
- ▶ With multiple endogenous variables, there are multiple first-stage regressions, and we should be wary if *any* of them suggests a weak instrument problem.
- ▶ Many have argued that the $F > 10$ cutoff is not stringent enough. See, for example, <https://arxiv.org/abs/2010.05058> Lee, McCrary, Moreira, and Porter

The model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

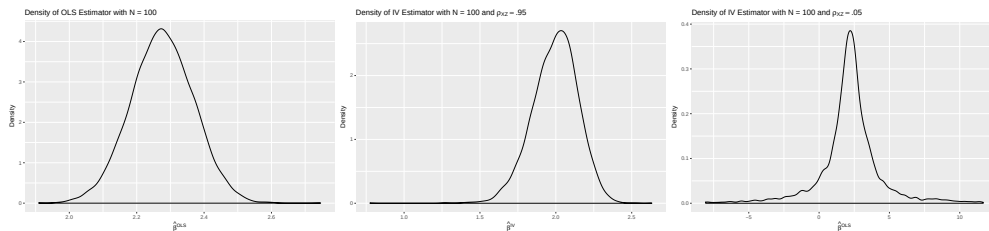
$$X_i = \pi_0 + \pi_1 Z_i^{(1)} + \pi_2 Z_i^{(2)} + \dots + \pi_m Z_i^{(m)} + \eta_i$$

- ▶ $\pi \approx 0 \Rightarrow$ because of *sampling error*, there is a good chance that $\hat{\pi} < 0$ even if $\pi > 0$
- ▶ If $\hat{X}$ is the wrong *sign* or close to zero then it is very likely that $\hat{\beta}^{IV}$ will behave poorly
- ▶ **NB:** As $N \rightarrow \infty$, $\hat{\pi} \rightarrow \pi$, so weak instruments are a *small sample problem*.

The Sampling Distribution of $\hat{\beta}$

$$Y = 1 + 2X + \varepsilon$$

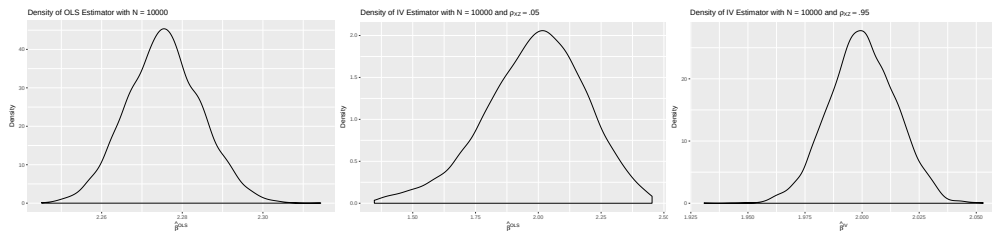
Figure 1: Distribution of $\hat{\beta}$ for OLS, strong IV, weak IV with $N = 100$



- ▶ OLS is biased with $\mu = 2.27$ and $\sigma^2 = .008$
- ▶ Strong IV is unbiased (asymptotically) with $\mu = 1.99$ and $\sigma^2 = .024$
- ▶ Weak IV is unbiased (asymptotically) with $\mu = 2.05$ but $\sigma^2 = \mathbf{1653.11}$

$$Y = 1 + 2X + \varepsilon$$

Figure 2: Distribution of $\hat{\beta}$ for OLS, weak IV, strong IV with $N = 10000$



- ▶ OLS remains biased with $\mu = 2.28$ and $\sigma^2 = .000085$
- ▶ Strong IV is consistent with $\mu = 1.99$ and $\sigma^2 = .00021$
- ▶ Weak IV is consistent with $\mu = 2.05$ and even with $N = 10k$, $\sigma^2 = .05$

What is the Danger of the CLT Failing?

- ▶ With weak IV, CLT fails and behaves poorly even if N is large
- ▶ Remember that for testing we want to reject the null hypothesis incorrectly 5% of the time
 - When the CLT fails, our trusty 1.96 critical value is way off base
 - Example rejection probabilities for test that $\hat{\beta} = 2$

Hypothetical Situation	$P(t > 1.96)$
Weak IV, $N = 100$	0.3%
Strong IV, $N = 100$	4.2%
Weak IV, $N = 10000$	3.2%
Strong IV, $N = 10000$	5.1%

- Goal should be 5% but for weak IV with N small, almost never reject; still only 3% even for N large—this is a bad test!

Example in R: Checking Angrist & Evans Validity

```
# First Stage Test in Angrist & Evans
flm1 = feols(morekids ~ samesex + black + hispan, data=babyData)
lh1 = linearHypothesis(flm1, "samesex = 0", test="F")

# First Stage Test in AE w/ Multiple Instruments
flm2 = feols(morekids ~ boyBoy + girlGirl + black + hispan, data=babyData)
lh2 = linearHypothesis(flm2, c("BoyBoy", "GirlGirl"), test="F")
```

Results:

- ▶ With “Same Sex”: $F = 25.874$
- ▶ With “BoyBoy” and “GirlGirl”: $F = 12.94$

Checking The IV Assumptions II

- ▶ The two major assumptions:
 1. Relevance: $Cov(X, Z) \neq 0$, or that $\mathbb{E}[\mathbf{z}_i \mathbf{x}_i']$ has full rank
 2. Exogeneity: All instruments must be uncorrelated with ε
- ▶ Now, let's talk about the exogeneity assumption.
- ▶ With one IV, it's impossible to test exogeneity, hence the need to evaluate exogeneity by thinking about whether some unobserved variables might be correlated with the instrument.
- ▶ Two popular tests:
 1. Hausman-Wu Test: Are $\hat{\beta}_{IV}$ and $\hat{\beta}_{OLS}$ the same?
 2. Sargan J Test: If we leave out z_i^* from our regression does $\mathbb{E}[(Y_i - X_i \hat{\beta}_{2SLS}) z_i^*] = 0$?
- ▶ These have fallen out of favor because they don't make sense if we have **heterogeneous treatment effects** (ie: β_i).
- ▶ Why? **different instruments** identify **different complier** groups.

- ▶ When the number of regressors equals the number of instruments, the model is **just identified**.
When there are more instruments than regressors, the model is **overidentified**.
- ▶ For a just identified model, the 2SLS estimator simplifies to

$$\mathbf{b}_{IV} = (\mathbf{z}'\mathbf{x})^{-1} \mathbf{z}'\mathbf{y},$$

which is often called simply the **IV estimator**.

- ▶ Recall the property of the OLS estimator:

$$\mathbf{x}'\mathbf{e}_{OLS} = \mathbf{0}, \text{ with } \mathbf{e}_{OLS} = (\mathbf{y} - \mathbf{x}\mathbf{b}_{OLS})$$

- ▶ Similarly, for the IV estimator,

$$\mathbf{x}'\mathbf{e}_{IV} = \mathbf{0}, \text{ with } \mathbf{e}_{IV} = (\mathbf{y} - \mathbf{x}\mathbf{b}_{IV})$$

- ▶ A consequence of this is that we can't test the assumption that the instrument(s) and error terms are uncorrelated by looking at the correlation between the instrument(s) and residuals.
- ▶ Similarly, the residuals from OLS don't allow us to test whether the error terms are correlated with the regressors.
- ▶ Stories about an IV being **excluded** are **just a story**.

Tests of Exogeneity?

One possibility is the **Durbin-Wu-Hausman Test** which compares $\Delta = \hat{\beta}_{OLS} - \hat{\beta}_{2SLS}$ under the null that X is **exogenous** (Z is assumed exogenous).

$H_0 : \beta_{OLS}, \beta_{2SLS}$ are consistent, β_{OLS} is efficient

$H_1 : \beta_{2SLS}$ is consistent, β_{OLS} is not .

This gives a chi-squared test statistic:

$$T = \Delta^T (V(\hat{\beta}_{2SLS}) - V(\hat{\beta}_{OLS}))^\dagger \Delta \sim \chi_K^2$$

This uses the **pseudo inverse** of the variance matrix where K is the rank of that matrix.

Idea: Reject H_0 if $\|\Delta\| \gg 0$ (2SLS and OLS disagree).

Tests of Exogeneity?

The other possibility is the **Sargan J-Test** which exploits **overidentification** (more instruments than endogenous regressors)

1. Pick an instrument(s) $z_i^* \in \mathbb{R}^{N \times L}$ to drop.
2. Run 2SLS using remainder except z_i^* to get $\hat{\beta}_{2SLS}^*$ with K restrictions.
3. Evaluate the **moment condition**: $g_n(\hat{\beta}_{2SLS}^*) = \mathbb{E}[(Y_i - X_i \hat{\beta}_{2SLS}^*) z_i^*]$.
4. Construct the quadratic form: $J = g_n(\hat{\beta}_{2SLS}^*)^T W g_n(\hat{\beta}_{2SLS}^*) \sim \chi^2_{L-K}$ where $L - K$ are the number of extra moments (dimension of z_i^*).

Software reports all three

```
> feols(worked ~ morekids + black + hisp + othrace + age + agefst |
+         samesex ~ boy1st + boy2nd, data=babydata)
TSLS estimation - Dep. Var.: worked
                  Endo.      : samesex
                  Instr.     : boy1st, boy2nd

Second stage: Dep. Var.: worked
Observations: 5,000
Standard-errors: IID

              Estimate Std. Error   t value   Pr(>|t|)
(Intercept)  0.645227   0.236454   2.728761 6.3795e-03 **
fit_samesex  -0.300863   0.403954  -0.744795 4.5643e-01
morekids     -0.128419   0.037575  -3.417631 6.3674e-04 ***
black        0.194002   0.032675   5.937304 3.0934e-09 ***
hisp         0.056280   0.045018   1.250171 2.1130e-01
othrace      0.072025   0.052196   1.379900 1.6768e-01
age          0.019904   0.003423   5.814619 6.4551e-09 ***
agefst       -0.025480   0.003559  -7.158666 9.3305e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.509524   Adj. R2: 0.043844
F-test (1st stage), samesex: stat = 3.21083, p = 0.040406, on 2 and 4,991 DoF.
Wu-Hausman: stat = 0.58365, p = 0.444923, on 1 and 4,991 DoF.
Sargan: stat = 0.32724, p = 0.567286, on 1 DoF.
```


1. We really want **strong instruments** that are good predictors of the **endogenous X 's**.
2. Everyone expects us to show the **first-stage F-statistic** telling us whether **excluded instruments** predict endogenous X **beyond the exogenous regressors**.
 - With appropriate clustering/standard errors we want to see $F > 10$. This is **large enough in practice** but probably **not in theory!**
 - Ideally we'd have a much stronger F -stat but good IV are hard to come by
3. We can't really test **exclusion** or **exogeneity** of your instruments.
 - This is largely a story that comes from some economic theory or understanding of the world.
 - There are tests but they don't show what we really care about
 - Wu-Hausman shows us if $\beta_{IV} = \beta_{OLS}$ (not if instruments are valid)
 - Sargan shows us if different instruments give the same $\hat{\beta}_{2SLS}$ (which might be because of **heterogeneous treatment effects**).

Simultaneous Equations

Simultaneous Equations Models

New Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$X_i = \alpha_0 + \alpha_1 Y_i + \nu_i$$

- ▶ In many economic environments, variables both determine each other “in equilibrium” or are determined together “jointly”
- ▶ A model where X and Y “cause” each other is called a **simultaneous equations model**
- ▶ Examples:
 - Angrist & Evans is an example of this: labor supply and fertility decisions are determined jointly, there is no one-way channel
 - Supply and demand is the canonical example: supply equations and demand equations are of interest but in equilibrium only observe $Q_S = Q_D$
 - Also common in macroeconomic contexts: interest rates, unemployment rates, etc.

Supply and Demand For Coffee, everything is linear

$$Q_t^d = \alpha_0 + \alpha_1 P_t + U_t$$

$$Q_t^s = \beta_0 + \beta_1 P_t + V_t$$

$$Q_t^d = Q_t^s$$

Solving for P_t, Q_t :

$$P_t = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{V_t - U_t}{\alpha_1 - \beta_1}$$

$$Q_t = \frac{\alpha_1 \beta_0 - \alpha_0 \beta_1}{\alpha_1 - \beta_1} + \frac{\alpha_1 V_t - \beta_1 U_t}{\alpha_1 - \beta_1}$$

Price is a function of both error terms, and we can't use a clever substitution to cancel things out.

To make things really obvious:

$$\begin{aligned}Cov(P_t, U_t) &= -\frac{Var(U_t)}{\alpha_1 - \beta_1} \\Cov(P_t, V_t) &= \frac{Var(V_t)}{\alpha_1 - \beta_1}\end{aligned}$$

- When demand slopes down ($\alpha_1 < 0$) and supply slopes up ($\beta_1 > 0$) then price is positively correlated with demand shifter U_t and negatively correlated with supply shifter V_t .

$$Cov(P_t, Q_t^d) = \alpha_1 Var P_t + Cov(P_t, U_t)$$

$$Cov(P_t, Q_t^s) = \beta_1 Var P_t + Cov(P_t, V_t)$$

► Bias in OLS estimate (Demand) $Bias(\alpha_1) = \frac{Cov(P_t, U_t)}{Var P_t}$.

► Bias in OLS estimate (Supply) $Bias(\beta_1) = \frac{Cov(P_t, V_t)}{Var P_t}$.

► We can actually write both this way when $Cov(U_t, V_t) = 0$:

$$\text{OLS Estimate} = \frac{\alpha_1 Var(V_t) + \beta_1 Var(U_t)}{Var(V_t) + Var(U_t)}$$

► More variation in supply $V_t \rightarrow$ better estimate of demand.

► More variation in demand $U_t \rightarrow$ better estimate of supply.

► Led Working to say the **statistical demand function** (OLS) is not informative about the economic demand function (or supply function).

IV Can Solve Simultaneity Bias

Why cover simultaneity bias *now*?

New Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$X_i = \alpha_0 + \alpha_1 Z_i + \alpha_2 Y_i + \nu_i$$

- ▶ If we have data on Z , we can potentially use Z as an instrument!
- ▶ What conditions would Z have to fulfill?
 1. Instrument Relevance: $\alpha_1 > 0$
 2. Instrument Exogeneity: $Cov(Z_i, \varepsilon_i) = 0$

Example: Angrist & Evans

Model:

$$\begin{aligned}Hours_i &= \beta_0 + \beta_1 Children_i + \varepsilon_i \\Children_i &= \alpha_0 + \alpha_1 Hours_i + \alpha_2 SameSex_i + \nu_i\end{aligned}$$

- ▶ *SameSex*: indicator for whether first two children have same sex
- ▶ Hours and children are determined jointly
- ▶ Also $Cov(\varepsilon, \nu)$ is unlikely to be 0
 - Education can matter for both number of children and hours worked (so education is part of ε and ν)
 - Being married may matter for both
 - Quality of non-financial benefits (e.g., maternity leave)
- ▶ So formally, the instrument *SameSex* is supposed to help solve simultaneity

Example: Angrist & Evans

TABLE 8—OLS AND 2SLS ESTIMATES OF LABOR-SUPPLY MODELS USING 1990 CENSUS DATA

	All women			Married women			Husbands of married women		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Estimation method	OLS	2SLS	2SLS	OLS	2SLS	2SLS	OLS	2SLS	2SLS
Instrument for <i>More than 2 children</i>	—	<i>Same sex</i>	<i>Two boys, Two girls</i>	—	<i>Same sex</i>	<i>Two boys, Two girls</i>	—	<i>Same sex</i>	<i>Two boys, Two girls</i>
Dependent variable: <i>Worked for pay</i>	−0.155 (0.002)	−0.092 (0.024)	−0.092 (0.024) [0.743]	−0.147 (0.002)	−0.104 (0.024)	−0.104 (0.024) [0.576]	−0.102 (0.001)	0.017 (0.009)	0.017 (0.009) [0.989]

Example: Porter (1983)

- ▶ The Joint Executive Committee's (19th Century railroad cartel) main business was transporting grain from Midwest to ports on East Coast.
- ▶ Porter (1983) estimates the following simultaneous system:

$$\text{demand: } \log Q_t = \alpha_0 + \alpha_1 \log P_t + \alpha_2 L_t + U_{1t}$$

$$\text{supply: } \log P_t = \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t}$$

where

- Q_t is the quantity (tonnage of grain) transported by the JEC
- P_t is the price per ton of grain transported
- L_t is a dummy for whether the Great Lakes were open to navigation
- S_t is a vector of time dummies
- I_t is an indicator for whether the JEC was colluding

Example: Porter (1983)

$$\text{demand: } \log Q_t = \alpha_0 + \alpha_1 \log P_t + \alpha_2 L_t + U_{1t}$$

$$\text{supply: } \log P_t = \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t}$$

$$\text{demand: } \log Q_t = \alpha_0 + \alpha_1 \log P_t + \alpha_2 L_t + U_{1t}$$

$$\text{supply: } \log P_t = \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t}$$

Example: Porter (1983)

TABLE 3 **Estimation Results***

Variable	Two Stage Least Squares (Employing <i>PO</i>)	
	Demand	Supply
<i>C</i>	9.169 (.184)	-3.944 (1.760)
<i>LAKES</i>	-.437 (.120)	
<i>GR</i>	-.742 (.121)	
<i>DM1</i>		-.201 (.055)
<i>DM2</i>		-.172 (.080)
<i>DM3</i>		-.322 (.064)
<i>DM4</i>		-.208 (.170)
<i>PO/PN</i>		.382

Example: Roberts and Schlenker (2013)

- ▶ How much does the biofuels mandate increase the price of grains? It depends on supply and demand.
- ▶ Roberts and Schlenker (2013) estimate the following supply and demand equations:

$$\text{demand: } \log Q_{dt} = \alpha_0 + \alpha_1 \log P_{dt} + \varepsilon_{dt}$$

$$\text{supply: } \log Q_{st} = \beta_0 + \beta_1 \log P_{st} + \beta_2 \omega_t + \varepsilon_{st}$$

- Q_{st} is global grain supply (calories)
- Q_{dt} is global grain consumption ($Q_{st} \neq Q_{dt}$ because of storage)
- The prices are measured at different times (planting vs harvest time)
- ω_t is a yield shock (due to weather), used as an instrument for demand
- ω_{t-1} is used as an instrument for supply (through storage, past production matters for current prices)

Example: Roberts and Schlenker (2013)

TABLE 1—SUPPLY AND DEMAND ELASTICITY (*FAO data*)

	Instrumental variables			Three-stage least squares		
	(1a)	(1b)	(1c)	(2a)	(2b)	(2c)
<i>Panel A. Supply equation</i>						
Supply elast. β_s	0.102*** (0.025)	0.096*** (0.025)	0.087*** (0.020)	0.116*** (0.019)	0.112*** (0.020)	0.097*** (0.019)
Shock ω_t	1.184*** (0.146)	1.229*** (0.138)	1.211*** (0.105)	1.249*** (0.111)	1.279*** (0.101)	1.241*** (0.091)
First stage ω_{t-1}	-3.901*** (1.145)	-3.628*** (0.945)	-3.824*** (0.910)	-3.546*** (0.800)	-3.113*** (0.704)	-3.226*** (0.731)
First stage ω_t	-2.918* (1.647)	-2.276* (1.294)	-2.372* (1.279)	-2.885*** (0.967)	-2.350*** (0.815)	-2.420*** (0.819)
<i>Panel B. Demand equation</i>						
Demand elast. β_d	-0.028 (0.021)	-0.055** (0.024)	-0.054** (0.022)	-0.034 (0.023)	-0.062*** (0.022)	-0.066*** (0.021)
First stage ω_t	-5.564*** (1.489)	-4.655*** (1.300)	-4.770*** (1.249)	-5.354*** (1.384)	-4.445*** (1.210)	-4.332*** (1.186)

From [Hughes, Knittel, Sperling \(2008\)](#)

As it turns out, identifying appropriate instrumental variables for gasoline demand is difficult. In this paper we experiment with crude oil production disruptions as instrumental variables.¹⁰ Disruptions are represented for three countries, Venezuela (VZ_{jt}), Iraq (IQ_{jt}) and the United States (US_{jt}). These three countries were selected because each has had its production of crude oil affected by external shocks that are unlikely to be related to gasoline demand shocks. In Venezuela, a strike by oil workers beginning in December 2002 cut production to near zero and has significantly affected output for several years. In Iraq, an international embargo and more recently war have caused major disruptions to oil operations. In the United States, production has been in steady decline since the 1970s due to declining resources. In 2005, hurricanes Katrina and Rita resulted in the temporary loss of several hundred thousand barrels of production in the Gulf of Mexico.

Simultaneity more examples

- ▶ **Download speeds and mobile data consumption:** Consumers use their smart phones more when download speeds are fast, but high usage leads to congestion, which slows down speeds.
- ▶ Road investment and travel demand: better roads are more appealing to travelers, and highly-used routes are likely to receive more funding .
- ▶ Public opinion influences policy, and policy influences public opinion (cigarettes, parental leave, same-sex marriage).
- ▶ Intensity of policing and crime rates.

Measurement Error

The problem: the X variable is measured with noise

- ▶ The idea mathematically:

$$\textbf{TRUTH: } Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\textbf{DATA: } Y = \beta_0 + \beta_1 X^* + \varepsilon^*$$

where X^* is a noisy measure of X

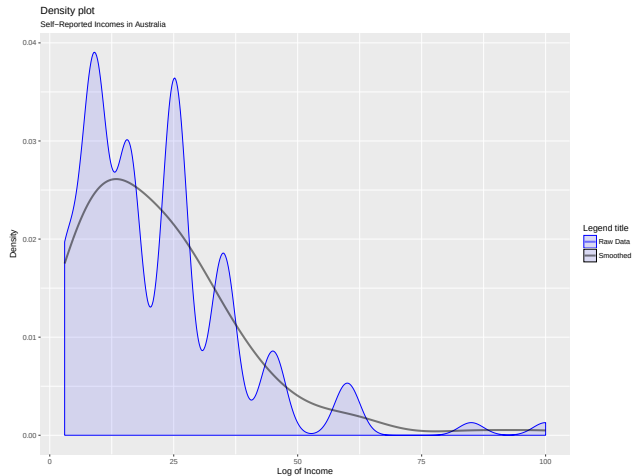
- ▶ Examples:

- Recording errors in data entry
- Recollection errors in survey data (these are frequent)
- Rounding errors
- Hard to measure variables (e.g., a firm's capital stock)

- ▶ Note: we can also have measurement error in Y . Turns out these behave differently

Visualizing Measurement Error: Self-Reported Australian Incomes

Survey question: What is your income in thousands?



Source: Kakadu Dataset

- Consider adding and subtracting noise to the truth:

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\&= \beta_0 + \beta_1 X_i^* + \underbrace{\beta_1 (X_i - X_i^*)}_{\text{"New" Error Term: } V_i} + \varepsilon_i \\&= \beta_0 + \beta_1 X_i^* + V_i\end{aligned}$$

- X_i^* and V_i likely to be correlated because X^* is part of V
- Extra assumptions:
 - **Classical Errors-in-Variables:**

$$X_i = X_i^* + u_i n^{-1} \sum_{i=1}^n m(y_i, \mathbf{x}_i; \theta)$$

Under classical measurement error:

$$Y_i = \beta_0 + \beta_1 X_i^* + \underbrace{\beta_1 u_i + \varepsilon_i}_{V_i}$$

► Using the population definition of $\hat{\beta}^{OLS}$:

$$\begin{aligned}\hat{\beta}^{OLS} &\rightarrow \frac{Cov(X^*, Y)}{Var(X^*)} \\ &= \frac{Cov(X + u, \beta_0 + \beta_1 X + \varepsilon)}{Var(X + u)} \\ &= \frac{Cov(X, \beta_0 + \beta_1 X + \varepsilon) + Cov(u, \beta_0 + \beta_1 X + \varepsilon)}{Var(X + u)} \\ &= \frac{\beta_1 Cov(X, X) + 0}{Var(X + u)}\end{aligned}$$

- Continuing from above:

$$\begin{aligned}\hat{\beta}^{OLS} &\rightarrow \frac{\beta_1 \text{Cov}(X, X) + 0}{\text{Var}(X + u)} \\ &= \beta_1 \times \frac{\text{Var}(X)}{\text{Var}(X + u)} \\ &= \beta_1 \times \frac{\sigma_X^2}{\underbrace{\sigma_X^2 + \sigma_u^2}_{\text{Attenuation}}}\end{aligned}$$

- Estimated coefficient converges to the truth times an attenuation term
- Attenuation term is less than 1 $\Rightarrow$ pushes coefficient towards zero
 - **NB:** It does *not* make the term more negative—it dampens the coefficient and preserves the sign
 - This “attenuates” the effect of X on Y . Hence the name: **attenuation bias**

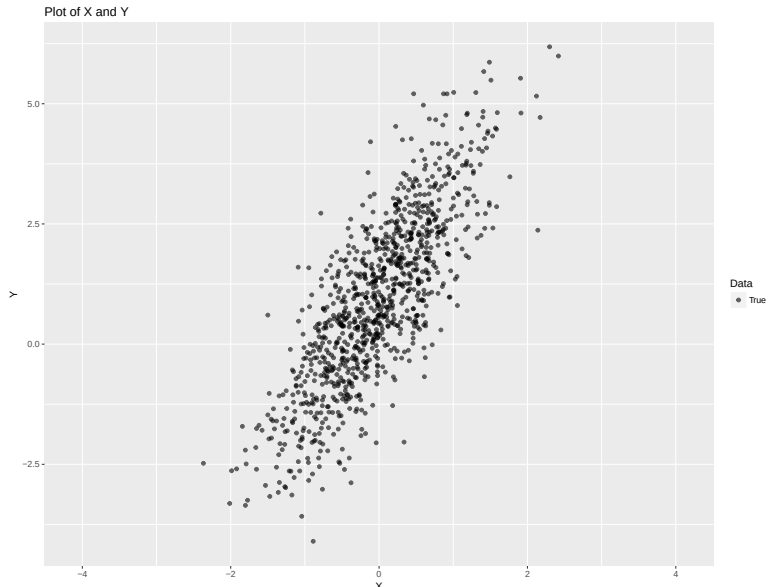
Measurement Error in X

- ▶ Why does measurement error attenuate the estimated effect of X on Y ?
 - Noise in measured X makes it more likely to see high X and low X with some Y because of randomness
 - Extreme example: fix X and keep adding noise—eventually X^* will look like noise itself
- ▶ Notice the following rearranging of the attenuation term:

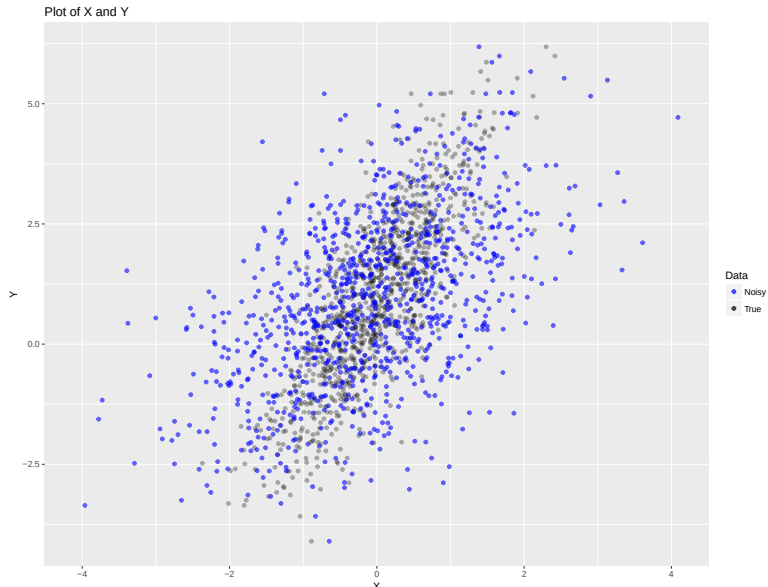
$$\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2} = \frac{1}{1 + (\sigma_u^2/\sigma_X^2)}$$

- σ_X^2/σ_u^2 is called the **signal-to-noise ratio**
- Larger signal to noise ratio $\Rightarrow$ smaller bias
- What matter is the *relative* variance of X to u
- If σ_u^2 is small *relative* to σ_X^2 then attenuation bias isn't too large

Visualizing Measurement Error



Visualizing Measurement Error



Visualizing Measurement Error



$$\begin{aligned}Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\&= \beta_0 + \beta_1 X_i^* + \underbrace{\beta_1 (X_i - X_i^*)}_{\text{"New" Error Term: } V_i} + \varepsilon_i \\&= \beta_0 + \beta_1 X_i^* + V_i\end{aligned}$$

- How do we address attenuation bias using instrumental variables?

- Suppose the econometrician observes $Y^* = Y + u$, where Y is the true value.

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\Y_i + Y_i^* &= \beta_0 + \beta_1 X_i + \varepsilon_i + Y_i^* \\Y_i^* &= \beta_0 + \beta_1 X_i + \varepsilon_i + (Y_i^* - Y_i)\end{aligned}$$

- Again, we can see this as changing the error term:

$$Y_i^* = \beta_0 + \beta_1 X_i + V_i$$

where $V_i = \varepsilon_i + (Y_i^* - Y_i)$.

- As long as this is classical measurement error, i.e. $\mathbb{E}[u \mid X] = 0$, then as long as exogeneity is satisfied with the original error term $\mathbb{E}[\varepsilon \mid X] = 0$, it will still be satisfied for the modified error term $\mathbb{E}[V \mid X] = 0$.

Generalized Method of Moments

IV as a method-of-moments estimator

- ▶ The Generalized Method of Moments includes everything we've done so far, and more.
- ▶ Recall how IV regression (just identified) results in residuals that satisfy

$$\mathbf{Z}' \mathbf{e} = \mathbf{0}.$$

This is the sample analog of the population moment $\mathbb{E} [\mathbf{z}_i e_i] = 0$.

- ▶ With a just identified model, the IV estimator is the unique estimator that sets $\mathbf{Z}' \mathbf{e} = \mathbf{0}$, just like how OLS is the unique estimator that gives $\mathbf{X}' \mathbf{e} = \mathbf{0}$
- ▶ Thus, we can derive the IV estimator from the starting point that we want to find parameters that give us residuals that are uncorrelated with our instruments.

- **Moments** are expectations of functions of the data:

$$\mathbb{E} [\mathbf{m} (y_i, \mathbf{x}_i; \boldsymbol{\theta}_0)] = 0,$$

where $\boldsymbol{\theta}_0$ is the true value of a vector of parameters of interest. Typically, we consider moments that are equal to zero – if they were set equal to something else, we could just subtract that from both sides.

- A crucial ingredient for the **method of moments** is that the sample analog of a moment,

$$n^{-1} \sum_{i=1}^n \mathbf{m}(y_i, \mathbf{x}_i; \boldsymbol{\theta})$$

will converge to the true analog, as long as $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ (law of large numbers)

- Furthermore, if the problem is set up well (i.e., if the model is **identified**), then

$$n^{-1} \sum_i^n \mathbf{m}(y_i, \mathbf{x}_i; \boldsymbol{\theta}) \nrightarrow 0$$

for any value of $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$.

$$n^{-1} \sum_i^n \mathbf{m}(y_i, \mathbf{x}_i; \boldsymbol{\theta}_0) \rightarrow 0, \text{ and } n^{-1} \sum_i^n \mathbf{m}(y_i, \mathbf{x}_i; \boldsymbol{\theta}) \not\rightarrow 0, \quad \forall \boldsymbol{\theta} \neq \boldsymbol{\theta}_0.$$

► Thus, let's try to find a parameter vector $\hat{\boldsymbol{\theta}}$ such that

$$n^{-1} \sum_i^n \mathbf{m}(y_i, \mathbf{x}_i; \hat{\boldsymbol{\theta}}) = 0.$$

- ▶ OLS can be derived as a method of moments estimator based on the moments

$$\mathbb{E} [\mathbf{x}_i \varepsilon_i] = 0$$

or

$$\mathbb{E} [\mathbf{x}_i (y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$$

- ▶ Similarly, IV estimation comes from the moments

$$\mathbb{E} [\mathbf{z}_i \varepsilon_i] = 0$$

or

$$\mathbb{E} [\mathbf{z}_i (y_i - \mathbf{x}_i' \boldsymbol{\beta})] = 0$$

- ▶ Econometrics II will cover GMM in more detail.